Marco Apolinario

Postdoctoral researcher – Memory-efficient on-device learning, brain-inspired AI, algorithm–hardware co-design

Department of Microelectronics,
Delft University of Technology,
Delft, The Netherlands
m.apolinariolainez@tudelft.nl
mapolinario94.github.io
Google Scholar

Ph.D. in Electrical and Computer Engineering with 7+ years of experience in machine learning research. I treat efficiency, locality, and continual adaptation as engineering constraints worth taking seriously, and design memory-efficient learning algorithms that scale to modern neural networks while running within the resource budgets of real on-device systems. Areas of focus include biologically-inspired local learning rules, continual and in-context learning, and algorithm–hardware co-design for energy-efficient, brain-inspired AI.

Research Interests

- Memory-efficient learning algorithms for on-device adaptation under hardware constraints.
- Continual learning and in-context adaptation in deep neural networks.
- Biologically-inspired local learning rules and gradient-free optimization.
- Algorithm–hardware co-design for energy-efficient, brain-inspired AI.

Education

Jan 2021 – Dec 2025 — **Ph.D. in Electrical and Computer Engineering**, *Purdue University*, West Lafayette, IN, USA.
Thesis advisor: Prof. Kaushik Roy. Research in hardware–software co-design for brain-inspired AI systems.

Mar 2013 – Dec 2017 — **B.S. in Electronics Engineering**, *National University of Engineering (UNI)*, Lima, Peru.
Focus on signal processing and machine learning. GPA: 3.5/4.0, ranked 3rd out of 28 students.

Fellowships and Awards

2026 — **Best Student Paper Award, WACV 2026**
Awarded for "Feedback Alignment Meets Low-Rank Manifolds – A Structured Recipe for Local Learning," algorithms track of IEEE/CVF WACV 2026.

2024 — **NSF AccelNet NeuroPAC Fellowship**
Selective international fellowship supporting cross-border collaborations in neuromorphic computing; awarded to conduct research on digital on-chip learning hardware accelerators at TU Delft.

2020 — **Graduate Peruvian Fellowship "Beca Generación del Bicentenario"**
Prestigious national fellowship fully funded by the Peruvian Ministry of Education; awarded to top scholars across Peru to pursue graduate studies abroad.

2017 — **"Julio Urbina Arias" Award**
Recognition for outstanding research and leadership contributions as an active member of the IEEE Student Branch, National University of Engineering, Lima, Peru.

Work Experience

Feb 2026 – Present — **Postdoctoral Researcher**, *Delft University of Technology (TU Delft)*, Delft, The Netherlands.
Research on hardware–algorithm co-design for NeuroAI.
Advisor: Prof. Charlotte Frenkel.

Aug 2021 – Dec 2025 — **Graduate Research Assistant**, *Purdue University – Nanoelectronics Research Laboratory (NRL)*, West Lafayette, IN, USA.
Research on neuro-inspired machine learning algorithms for emerging hardware technologies, with emphasis on scalability and energy efficiency in neuromorphic systems.
Advisor: Prof. Kaushik Roy.

Sep 2024 – Dec 2024 **Visiting Researcher**, *Delft University of Technology (TU Delft)*, Delft, The Netherlands.
Three-month, fellowship-funded research visit (NSF AccelNet NeuroPAC Fellowship); research on hardware–algorithm co-design for on-device learning using local learning rules.
Advisor: Prof. Charlotte Frenkel.

May 2023 – Aug 2023 **Systems Engineering Intern**, *Texas Instruments – Kilby Labs*, Dallas, TX, USA.
Hardware-aware neural architecture and quantization search via evolutionary optimization for low-power deployment; achieved 10× reduction in model search time and 5% performance improvement on keyword spotting.

Jul 2017 – Dec 2020 **Research Assistant in Computer Vision**, *INICTEL-UNI (National Institute for Research and Training in Telecommunications)*, Lima, Peru.
Energy-efficient ML for natural-resource monitoring (timber identification, underwater acoustic inversion, satellite cloud segmentation, river-level estimation); deployment on low-power embedded systems for precision agriculture.

Publications

*A separate document containing my five most representative publications and the reason for selecting each is provided alongside this CV. The candidate's name is shown in **bold**.*

2026 **Feedback Alignment Meets Low-Rank Manifolds: A Structured Recipe for Local Learning** [Best Student Paper Award]
A. Roy, **M. Apolinario**, S. Das Biswas, K. Roy. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

2026 **LANCE: Low Rank Activation Compression for Efficient On-Device Continual Learning**
M. Apolinario, K. Roy. *arXiv preprint*.

2025 **CODE-CL: Conceptor-Based Gradient Projection for Deep Continual Learning**
Marco Apolinario, Sakshi Choudhary, Kaushik Roy. *International Conference on Computer Vision (ICCV)*.

2025 **TESS: A Scalable Temporally and Spatially Local Learning Rule for Spiking Neural Networks**
Marco Apolinario, Kaushik Roy, Charlotte Frenkel. *International Joint Conference on Neural Networks (IJCNN)*.

2025 **LLS: Local Learning Rule for Deep Neural Networks Inspired by Neural Activity Synchronization**
Marco Apolinario, Arani Roy, Kaushik Roy. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

2025 **S-TLLR: STDP-inspired Temporal Local Learning Rule for Spiking Neural Networks**
Marco Apolinario, Kaushik Roy. *Transactions on Machine Learning Research (TMLR)*.

2024 **Unearthing the Potential of Spiking Neural Networks**
Sayeed Chowdhury, Adarsh Kosta, Deepika Sharma, **Marco Apolinario**, Kaushik Roy. *Design, Automation & Test in Europe Conference & Exhibition (DATE)*.

2024 **HALSIE: Hybrid Approach to Learning Segmentation by Simultaneously Exploiting Image and Event Modalities**
Shristi Das Biswas, Adarsh Kosta, Chamika Liyanagedera, **Marco Apolinario**, Kaushik Roy. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

2023 **Hardware/Software Co-Design With ADC-Less In-Memory Computing Hardware for Spiking Neural Networks**
Marco Apolinario, Adarsh Kosta, Utkarsh Saxena, Kaushik Roy. *IEEE Transactions on Emerging Topics in Computing*.

2023 **Live Demonstration: ANN vs SNN vs Hybrid Architectures for Event-based Real-time Gesture Recognition and Optical Flow Estimation**
Adarsh Kosta, **Marco Apolinario**, Kaushik Roy. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*.

2020 **Method of Estimating River Levels with Reflective Tapes Using Artificial Vision Techniques**
Lidia Lopez, **Marco Apolinario**, Samuel Huaman. *Proceedings of the 5th Brazilian Technology Symposium*.

2019 **Open Set Recognition of Timber Species Using Deep Learning for Embedded Systems**
Marco Apolinario, Daniel Urcia, Samuel Huaman. *IEEE Latin America Transactions*.

2019 **Estimation of 2D Velocity Model using Acoustic Signals and Convolutional Neural Networks**
Marco Apolinario, Samuel Huaman, Giorgio Morales, Daniel Diaz. *IEEE International Conference on Electronics, Electrical Engineering and Computing (INTERCON)*.

2018 **Deep Learning Applied to Identification of Commercial Timber Species from Peru**
Marco Apolinario, Samuel Huaman, Gabriel Orellana. *IEEE International Conference on Electronics, Electrical Engineering and Computing (INTERCON)*.

Registered Software / Intellectual Property

Software products co-developed at INICTEL-UNI and registered with INDECOPI, Peru's national authority for intellectual property.

Feb 2022 **SM-GROVER –** Geo-referenced monitoring system for COVID-19 with virtual assistant and respiratory-rate estimation. Software IP registration, INDECOPI 00376-2022. Co-developer.

Jun 2021 **SINIM-1 –** Cloud identification in multispectral satellite imagery. Software IP registration, INDECOPI 00740-2021. Co-developer.

Mar 2021 **KASPI –** Mobile application for identification of Peruvian timber species. Software IP registration, INDECOPI 00294-2021. Co-developer.

Teaching

Jan 2020 **Instructor, Short Course on Digital Image Processing** (24 hours), *INICTEL-UNI*, Lima, Peru.
Designed and delivered a 24-hour short course on digital image processing for early-career engineers and students.

2015 – 2017 **Workshop instructor, IEEE Student Branch**, *National University of Engineering (UNI)*, Lima, Peru.
Taught hands-on workshops on signal and image processing and introductory machine learning for undergraduate audiences as part of the IEEE Signal Processing Society and Robotics & Automation Society student chapters.

Mentorship and Outreach

2025 – Present **Mentor, Talento Guía (PRONABEC, Peru National Scholarship Program).** Mentor for incoming Peruvian graduate students abroad (Beca Generación del Bicentenario Fellows), providing academic, cultural, and professional guidance for their transition to international graduate programs.

2021 – 2022 **Mentor, Serendipity: Mentorship in Science Program.** Mentored Peruvian undergraduate students in STEM on career development, research opportunities, and pathways to graduate school.

2020 – Present **Speaker – STEM Outreach.** Invited speaker at IEEE and university events in Peru, promoting undergraduate research and STEM engagement.

2016 – 2017 **Vice Chair, IEEE Signal Processing Society Student Chapter, UNI (Peru).** Co-founded the first IEEE Signal Processing Society Student Chapter in Peru; promoted undergraduate-led research projects in signal and image processing.

2015 – 2016 **Research Director, IEEE Robotics and Automation Society Student Chapter, UNI (Peru).** Led student research groups in robotics and computer vision; organized national robotics competitions, technical talks, and workshops.

Travel Grants

2025 **LatinX in AI (LXAI) Travel Grant – NeurIPS**, San Diego, CA, USA.
Competitively awarded grant supporting participation and presentation at the LatinX in AI Research Workshop co-located with NeurIPS.

2025 **ICCV Broadening Participation Travel Grant**, Honolulu, HI, USA.
Awarded for conference participation based on scientific contribution, need, and commitment to broadening participation within the ICCV community.

Invited Talks

Apr 2025 **Energy-Efficient Brain-Inspired Learning in Deep Neural Networks**. IEEE Signal Processing Society student chapter, National University of Engineering, Lima, Peru (online).

Nov 2024 **Local Learning for Deep Neural Networks**. Data Management and Biometrics Group, University of Twente, The Netherlands.

Jul 2024 **Neuromodulation on Brain and Machines** (co-presented with Dr. Kathryn Simone). Telluride Neuromorphic Cognition Engineering Workshop, Telluride, CO, USA.

Jun 2024 **Hardware/Software Co-design with ADC-Less In-Memory Computing for SNNs**. In-Memory Computing Applications Workshop at DAC 2024, San Francisco, CA, USA.

Aug 2023 **Enabling High-Performance ADC-Less In-Memory Computing for Deploying SNNs Through Hardware-Aware Training**. Workshop on Modeling & Simulation of Systems and Applications (ModSim23), Seattle, WA, USA.

Jul 2021 **Experiencia estudiando un posgrado en EEUU – Becas y oportunidades**. IEEE Signal Processing Society student chapter, National University of Engineering, Lima, Peru (online).

Academic Service

Review Comm. Member (RCM) IEEE International Symposium on Circuits and Systems (ISCAS), 2025 – coordinated the peer-review process for three papers, synthesizing feedback from multiple reviewers and submitting preliminary decisions (role analogous to Associate Editor / Area Chair).

Reviewer (journals) IEEE Transactions on Image Processing (TIP); IEEE Transactions on Cognitive and Developmental Systems (TCDS); IEEE Transactions on Biomedical Circuits and Systems (TBioCAS); IEEE Latin America Transactions.

Reviewer (conferences) NeurIPS (2024, 2025); ICLR (2025, 2026); ICML (2025); ICCV (2025); ECCV (2026); AAAI (2026); CVPR (2025); IJCNN (2025); ICANN (2024, 2025); IEEE INTERCON (2024).

In the Press

2026 Interview in **APTC *La Revista*** (magazine of the Peruvian Telecommunications Association): "El cerebro humano es la computadora más eficiente" ["The human brain is the most efficient computer"].

Professional Development

2024 **Telluride Neuromorphic Cognition Engineering Workshop**
Highly selective international workshop; hands-on projects on neuromodulation mechanisms for synaptic plasticity and reinforcement learning.

2024 **SRC TECHCON**
Semiconductor Research Corporation's flagship annual conference, Austin, TX, engaging with industry leaders on advances in semiconductor and AI research.

2021	**Neuromatch Academy – Computational Neuroscience Summer School** Intensive online summer school on computational neuroscience, exploring parallels between artificial neural networks and in-vivo brain responses to visual stimuli.

Skills

Programming & HDLs	Python, C/C++, VHDL/Verilog, Git.
EDA Tools	Cadence Virtuoso, Quartus Prime, Eagle PCB.
ML Frameworks	PyTorch, TensorFlow/Keras.
Languages	English (fluent), Spanish (native).

Relevant Coursework

Graduate Electronics	AI Hardware; Computer Architecture; System-on-Chip Design; Analog CMOS Design; Advanced VLSI Design; MOS VLSI Design; Solid State Devices.
Graduate Computer Science	Applied Quantum Computing; Optimization for Deep Learning; Computational Methods in Optimization; Artificial Intelligence.